

ERWIN PAR-T

A Tree-based Hierarchical Transformer for Large Scale Physical Systems (Possibly for Jet Tagging)

05/21/26

Standard Self Attention Mechanism

$$Q, K, V = XW_q, XW_k, XW_v, \quad \text{Att}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{C'}} + B \right) V$$

- Particle Attention Block :

$$P - \text{MHA}(Q, K, V) = \text{SoftMax}(QK^T / \sqrt{d^K} + U)V$$

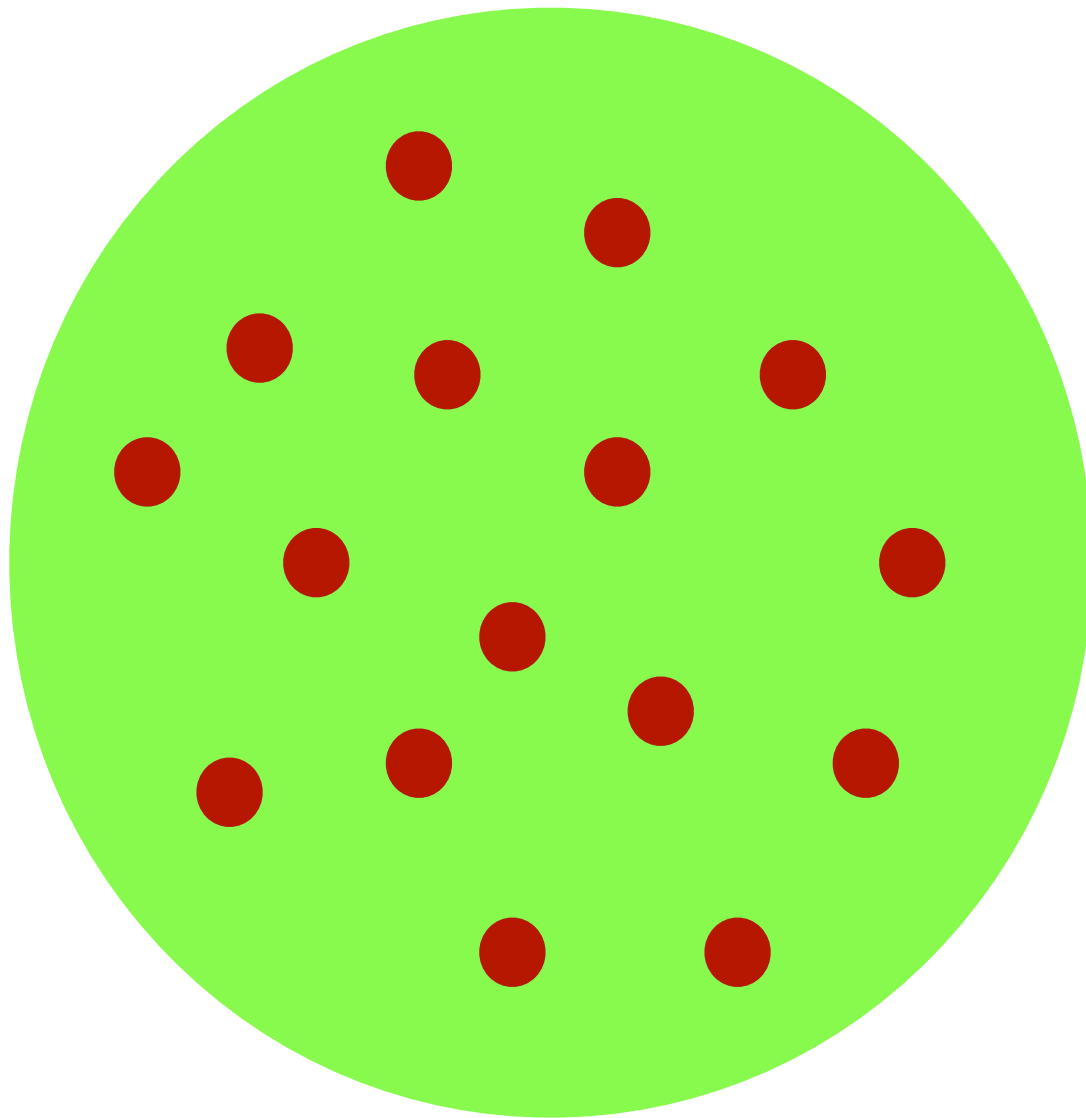
Where U is pre-softmax attention weight to incorporate particle attention feature.

- Without any locality constraints, computes interactions between all elements in the input set.
- In result, we get a quadratic computational cost $\mathcal{O}(N^2)$

Ball and Ball Tree in a Point Cloud

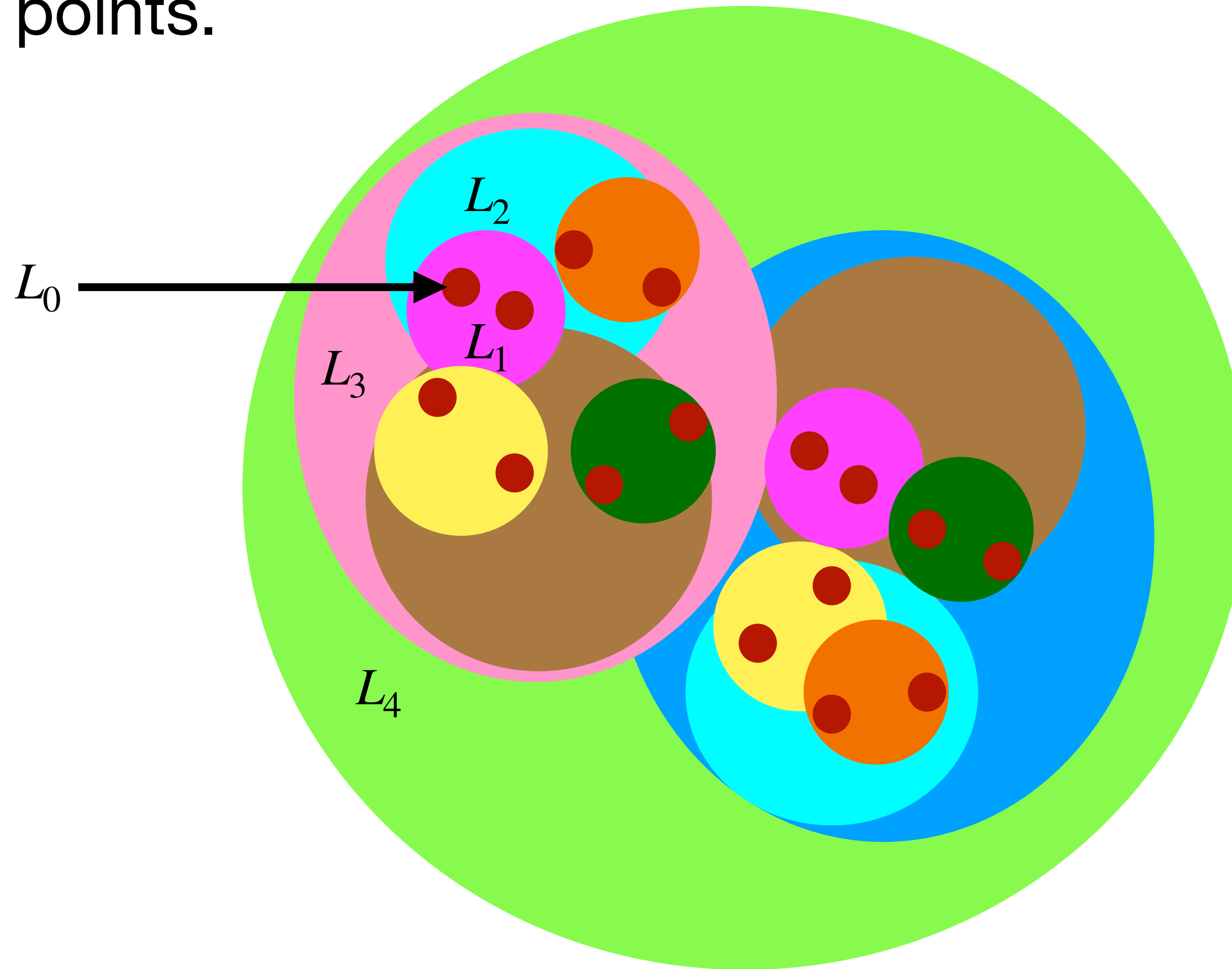
- A ball is a region bounded by a hypersphere in $\mathbb{R}^d$.
- Each ball is represented by the coordinates of its center $\mathbf{c}$ and radius r .

$$B = B(c, r) = \left\{ z \in \mathbb{R}^d \mid \|z - c\|_2 \leq r \right\}$$



Ball and Ball Tree in a Point Cloud

- A ball tree T on a point set P is a hierarchical sequence of partitions $\{L_0, L_1, \dots, L_m\}$.
- Each Level L_i cover disjoint balls that covers P , at leaf level ($i = 0$) the nodes are the original points.

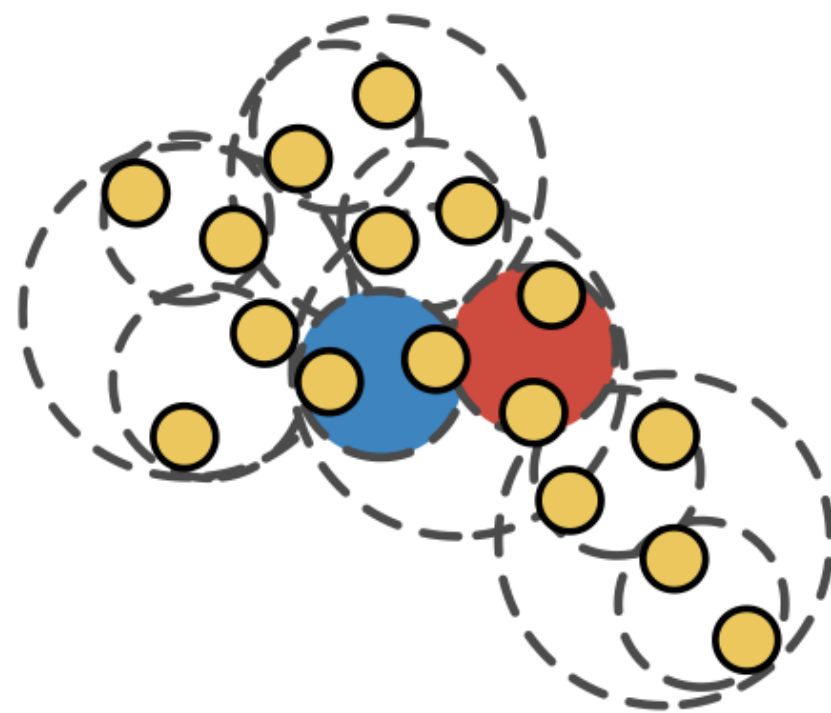


Ball Tree Attention

- For each ball attention operator we specify a level k of the ball tree where each ball $B \in L_k$ contains 2^k leaf nodes.
- That means larger balls will capture long range dependencies while smaller balls are more resources efficient.

$$leaves_B = \{B' \in L_{leaf}\}$$

Ball tree



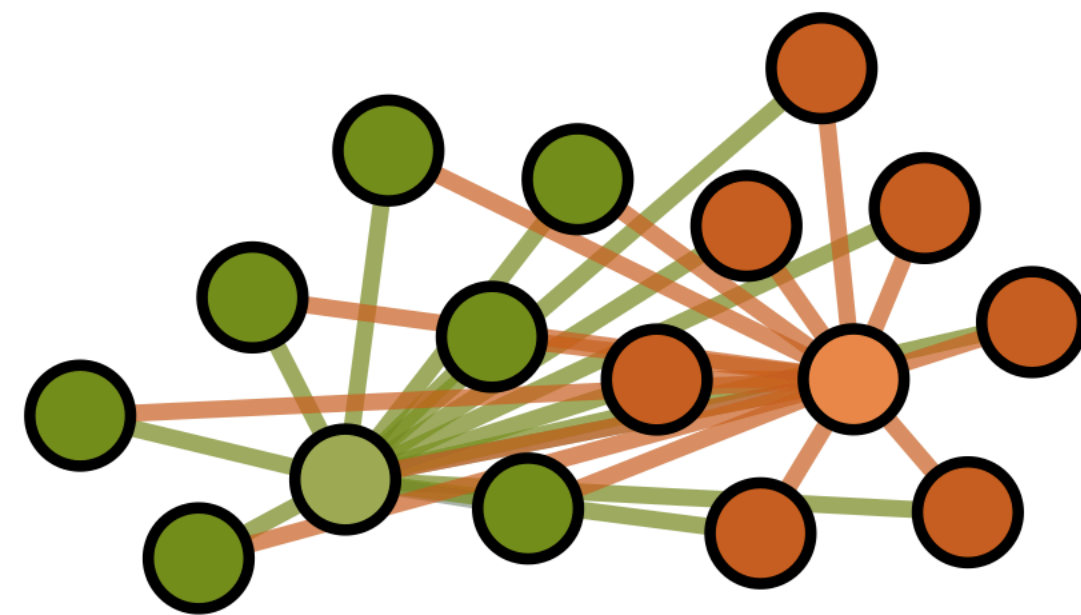
$$X_B = \{x'_B \in X_{leaf} \mid B' \in leaves_B\}$$

$$X'_B = BAtt(X_B) = Att(X_B W_q, X_B W_k, X_B W_v)$$

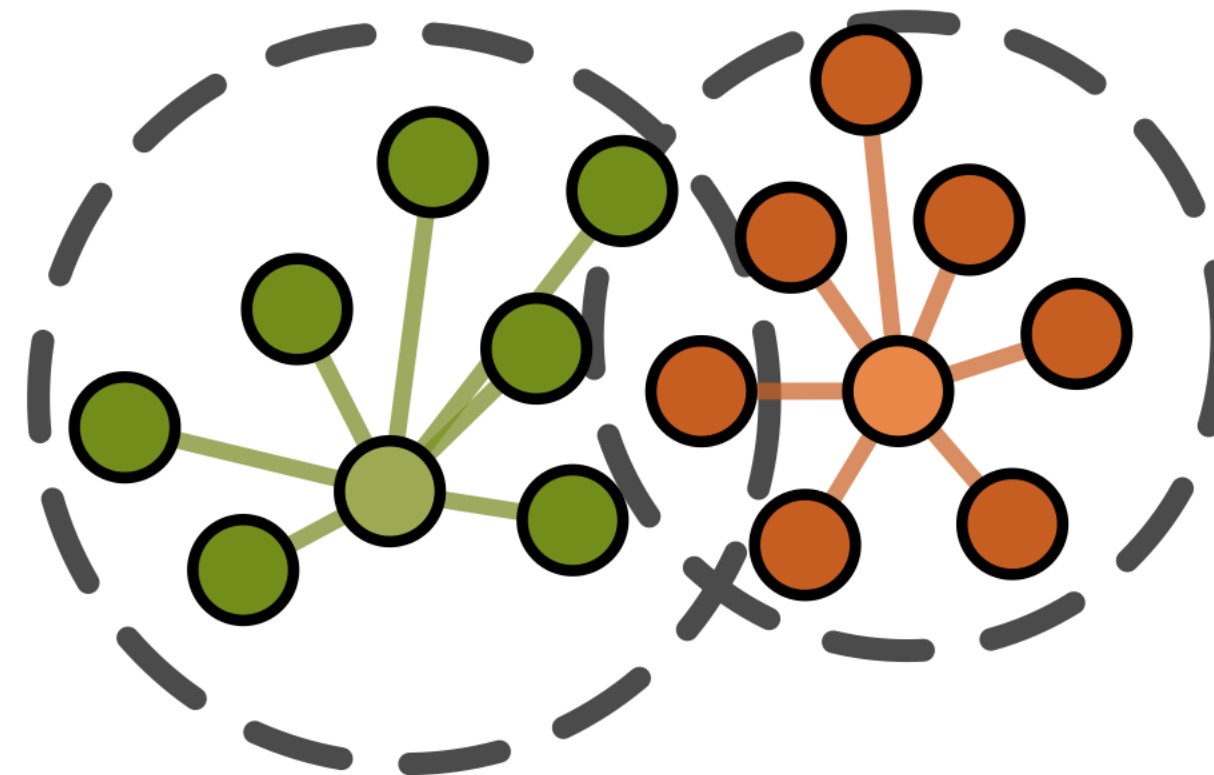
Computational Cost

- Now, as computation for attention is done independently for each ball. The computational cost reduced from quadratic to linear i.e $\mathcal{O}(B^2 \cdot \frac{n}{|B|})$.
Quadratic in ball size and linear in no of balls.

Attention $\mathcal{O}(n^2)$



Ball Attention $\mathcal{O}(n)$



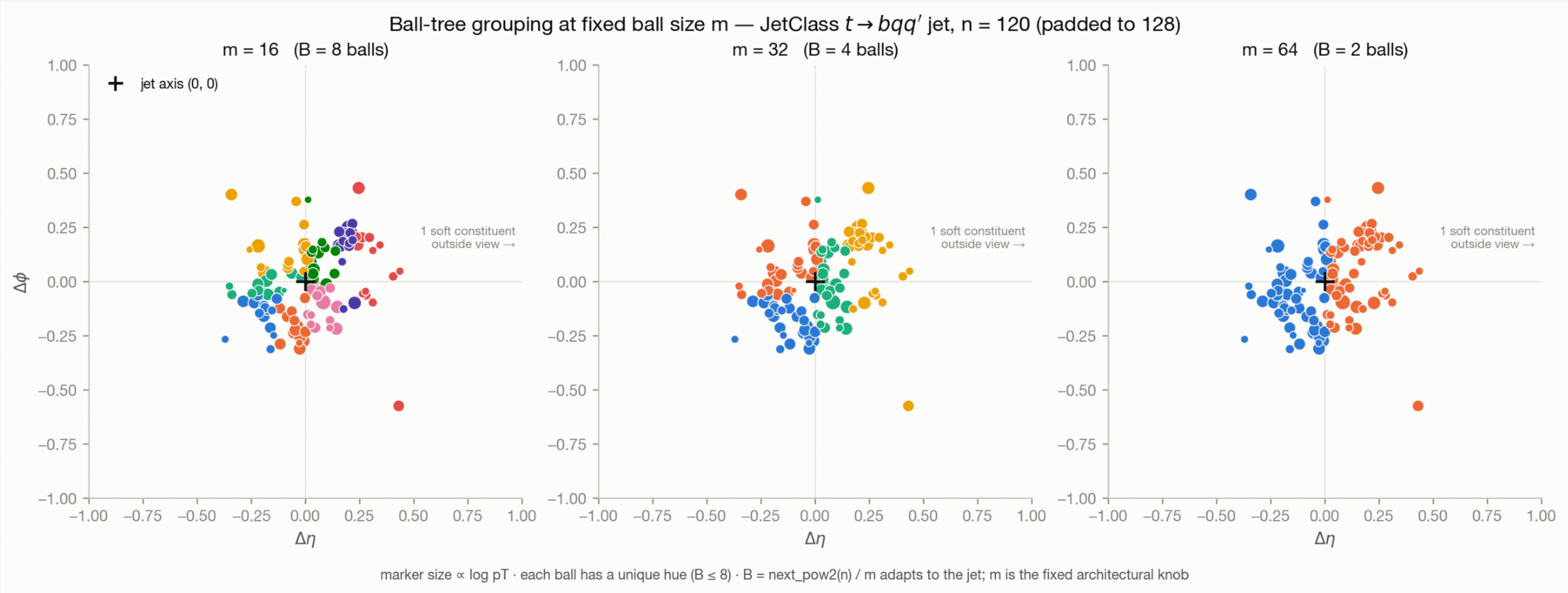
ErwinParT with JetClass dataset.

- I made the Ball attention for our efficient particle transformer.
- Ran it with only 10% of JetClass dataset with RTX 4090

Model	Accuracy	AUC	H→bb ⁻ Rej 50%	H→cc ⁻ Rej 50%	H→gg Rej 50%	H→4q Rej 50%	H→ℓvqq' Rej 99%	t→bqq' Rej 50%	t→bℓv Rej 99.5%	W→qq' Rej 50%	Z→qq ⁻ Rej 50%
PFN	0.772	0.9714	2924	841	75	198	265	797	721	189	159
P-CNN	0.809	0.9789	4890	1276	88	474	947	2907	2304	241	204
ParticleNet	0.844	0.9849	7634	2475	104	954	3339	10526	11173	347	283
ParT (plain)	0.849	0.9859	9569	2911	112	1185	3868	17699	12987	384	311
ErwinParT	0.825	0.9819	8000	1681	97	590	1695	5556	4545	296	244
ParT	0.861	0.9877	10638	4149	123	1864	5479	32787	15873	543	402

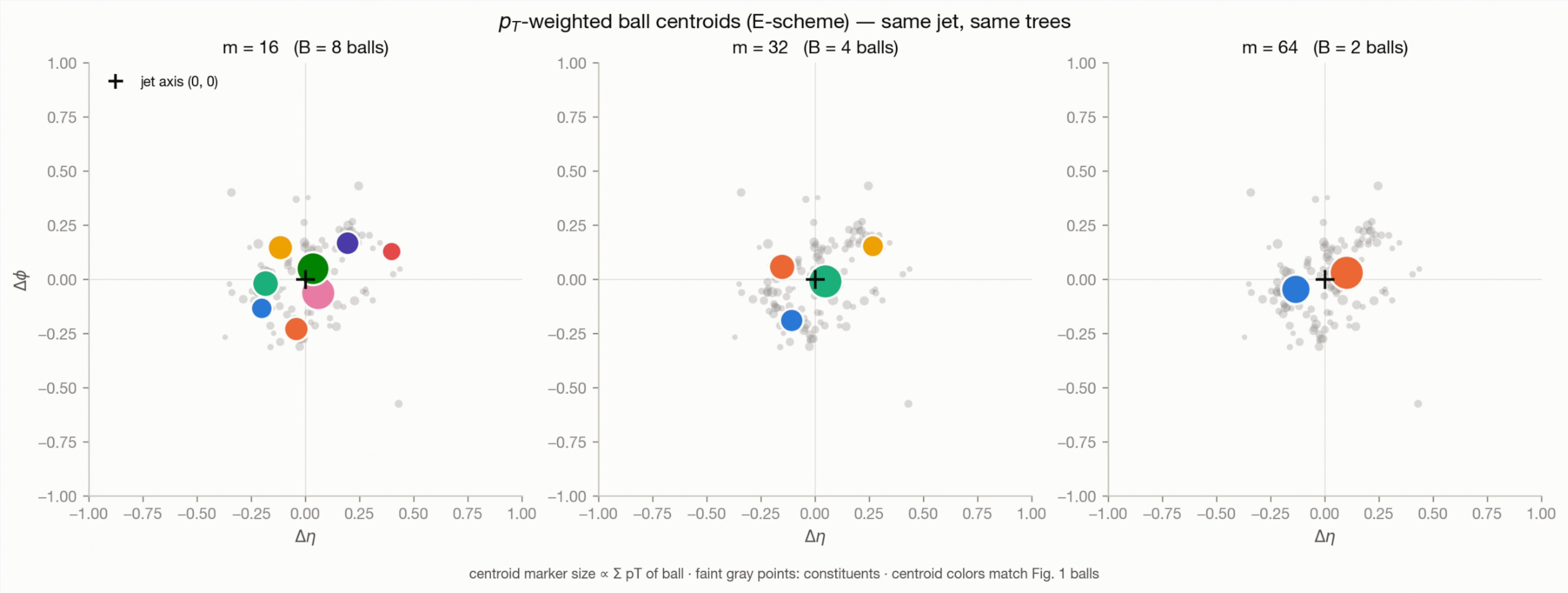
How the tree groups constituents in ($\Delta\eta$, $\Delta\phi$)

carried over from the v1 ball-tree study · single level, ball size $m \in \{16, 32, 64\}$, jets padded to 128



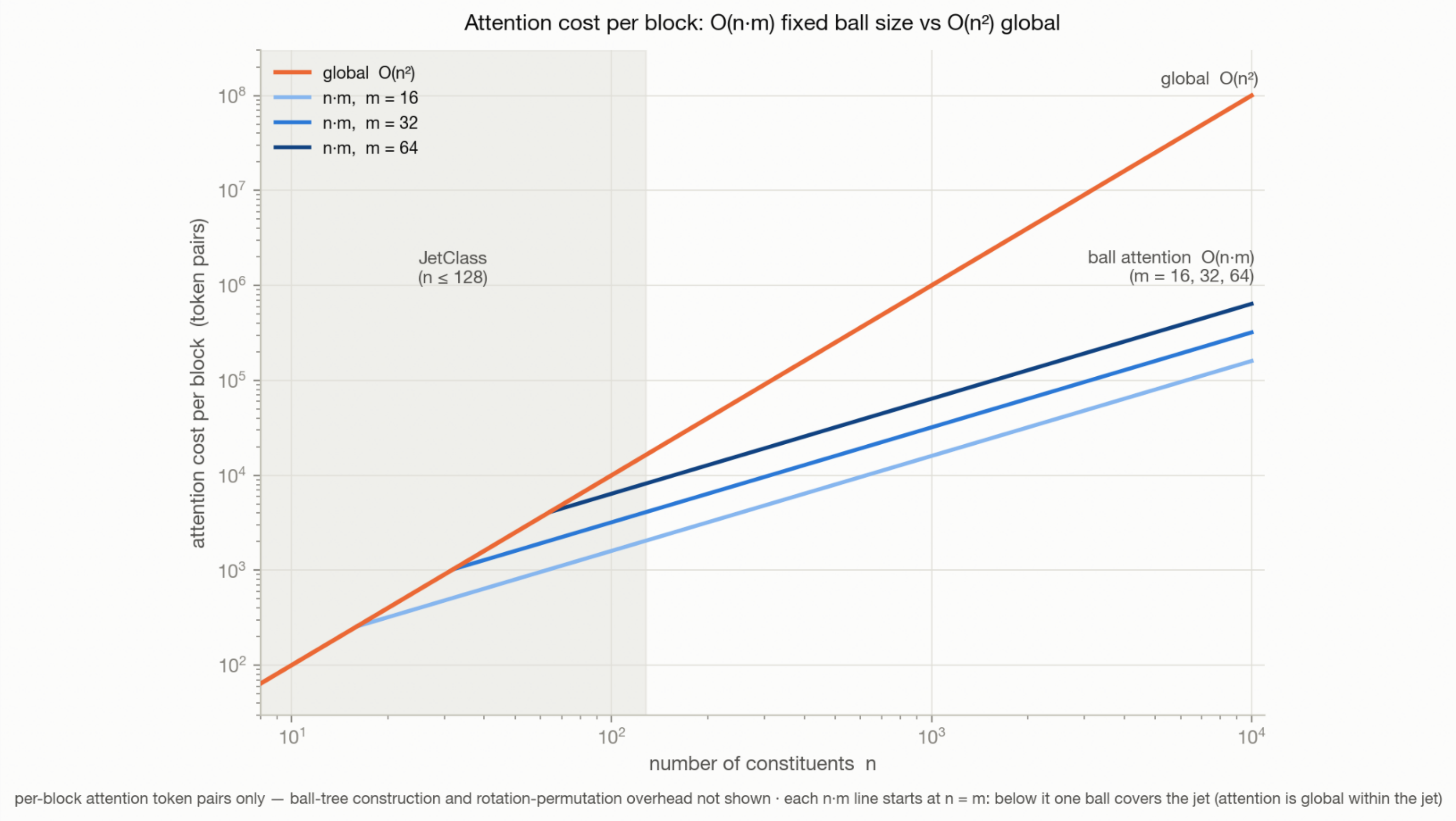
p_T -weighted ball centroids

carried over from the v1 ball-tree study · single level, ball size $m \in \{16, 32, 64\}$, jets padded to 128



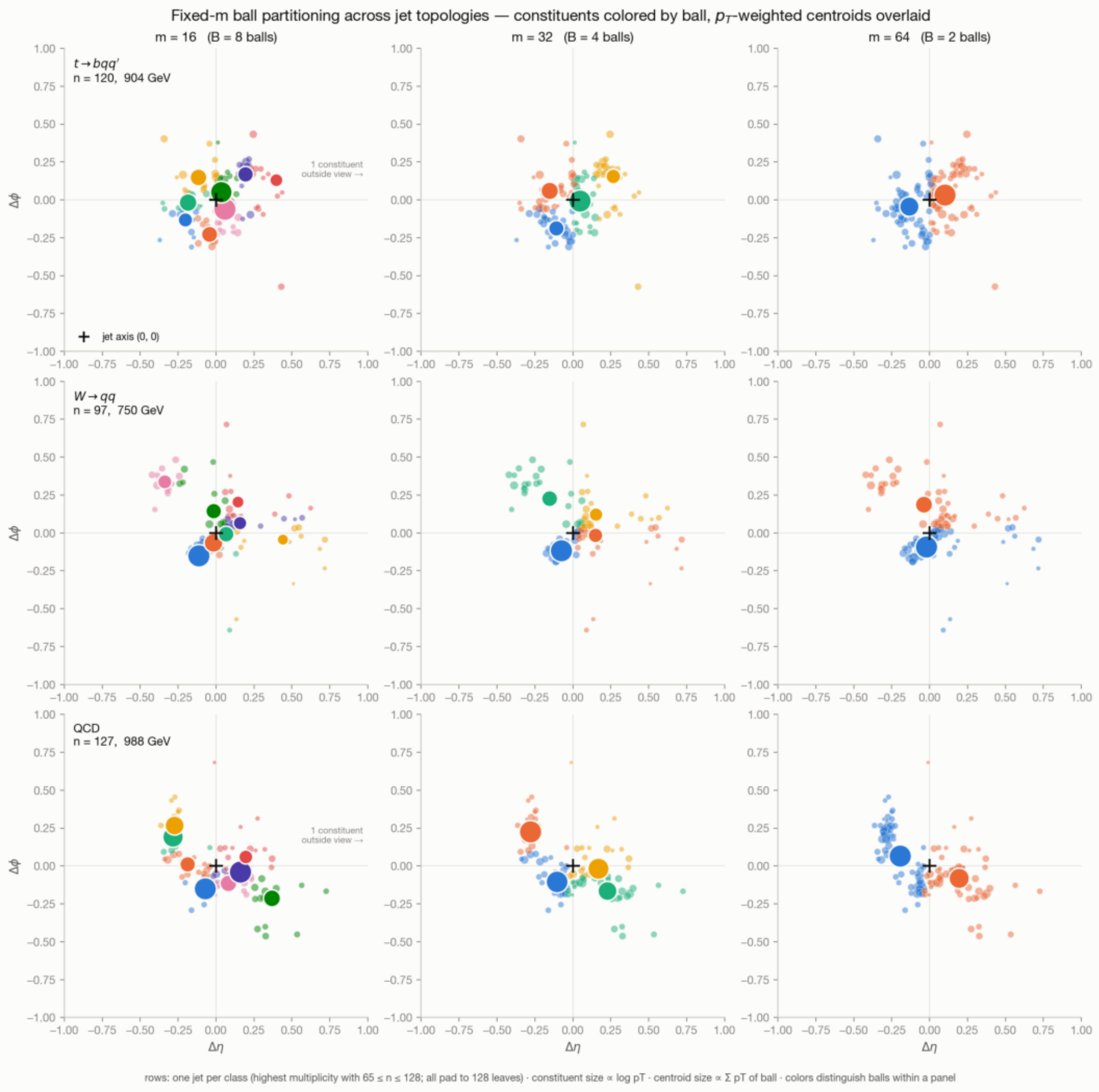
O(n·m) against O(n²)

carried over from the v1 ball-tree study · single level, ball size $m \in \{16, 32, 64\}$, jets padded to 128



The same partitioning on top, W and QCD jets

carried over from the v1 ball-tree study · single level, ball size $m \in \{16, 32, 64\}$, jets padded to 128



Ball-tree attention

for jet tagging

Does a jet need every constituent
to see every other one?

ErwinParTv2

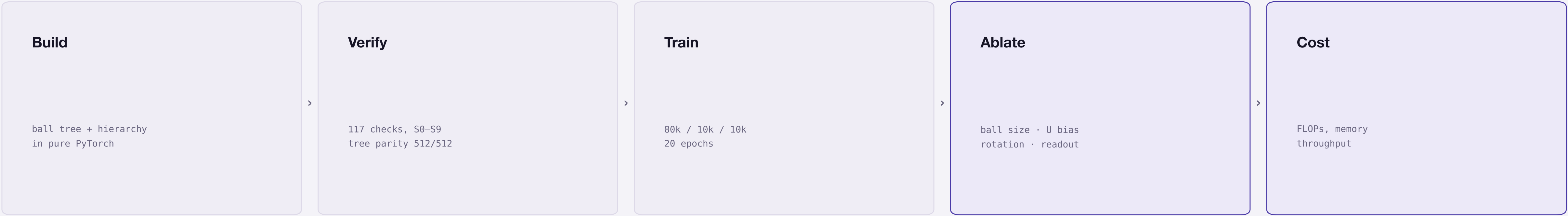
JetClass · 100k jets

117 verification checks

2 seeds

No ParT accuracy baseline: ParT could not be trained on this hardware. Every accuracy number here is the model against itself.

The analysis in one line

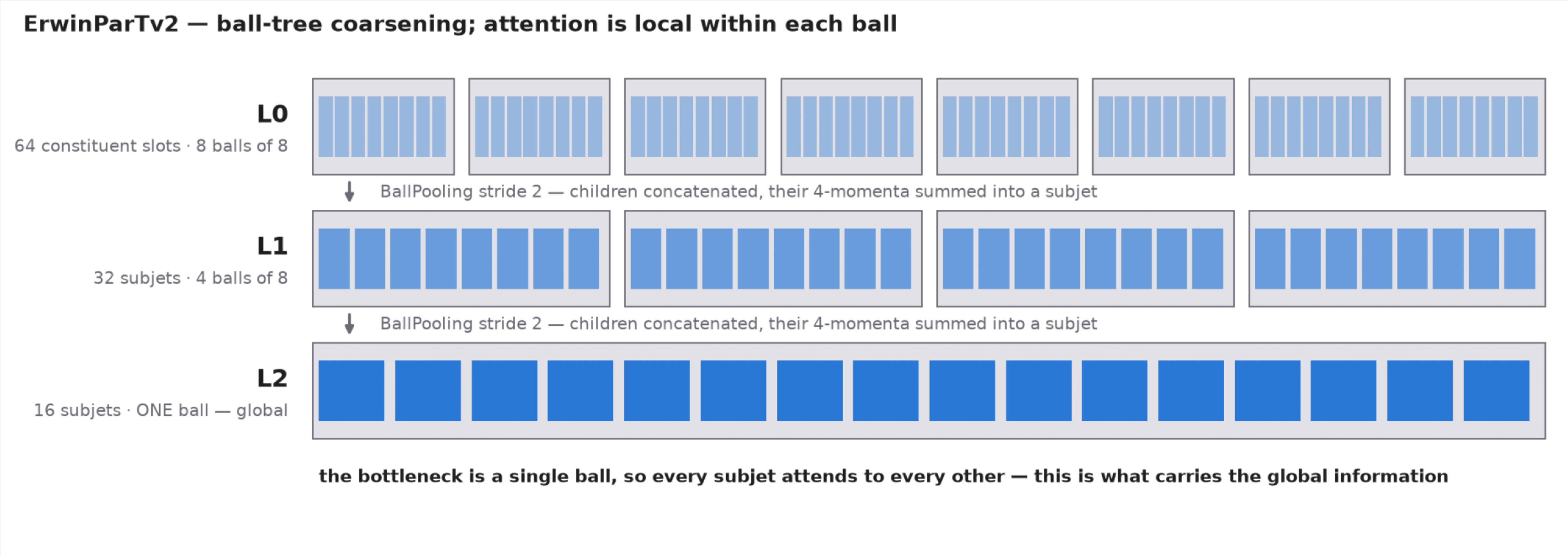


slides 14–16 – what it is, and is the input right

slides 17–20 – does locality cost anything

slide 21 – what it buys

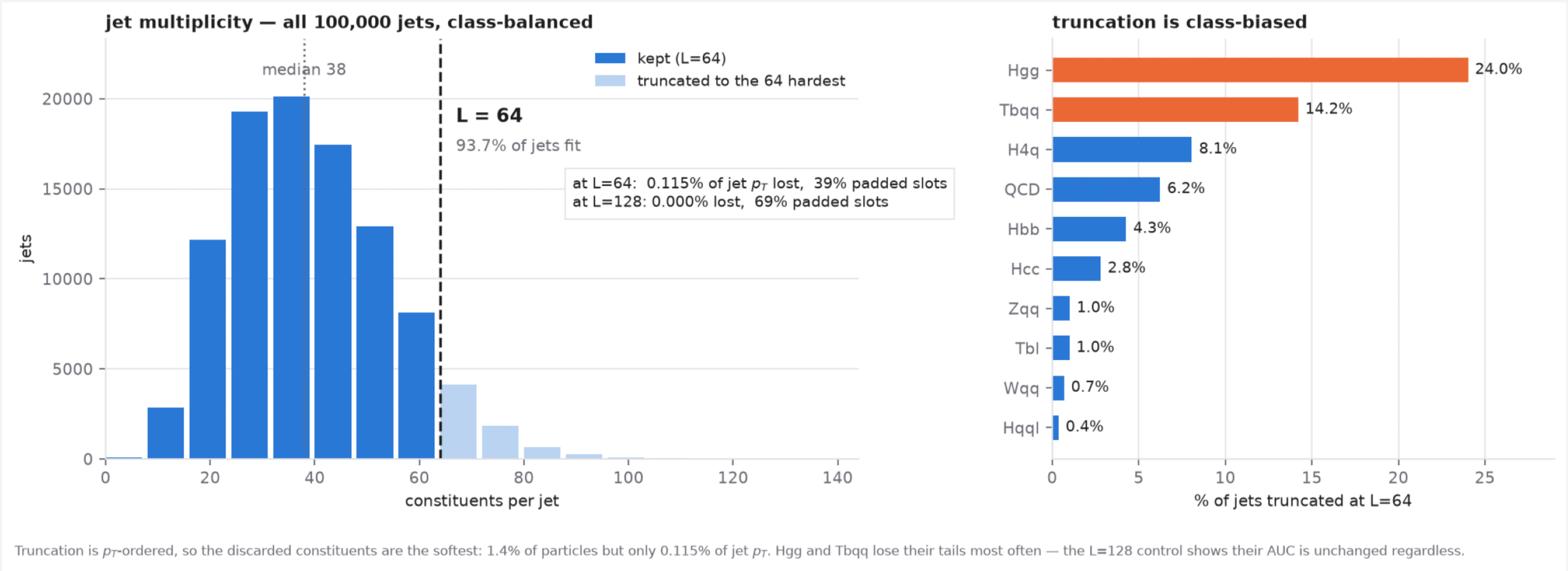
Attention is local; the hierarchy carries global information



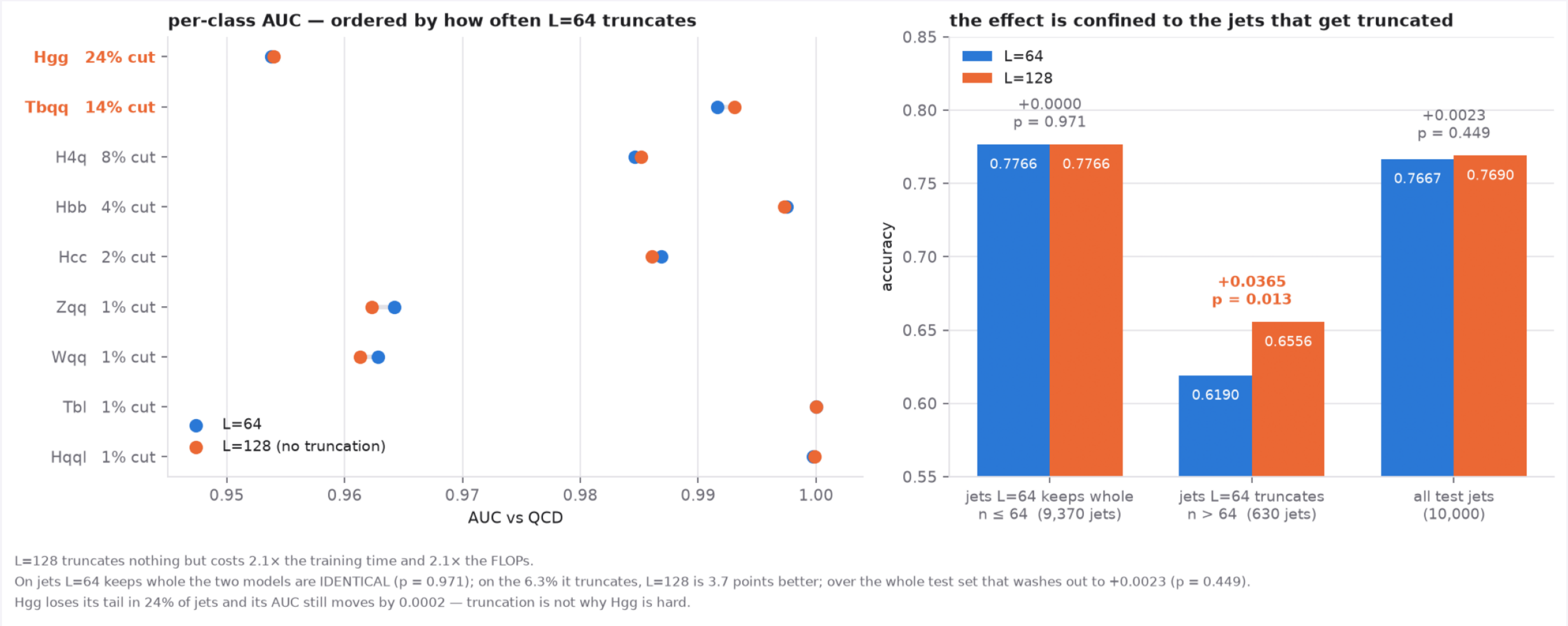
64 > 32 > 16 nodes

bottleneck = one ball

Why 64 slots per jet



Does truncation cost us anything?



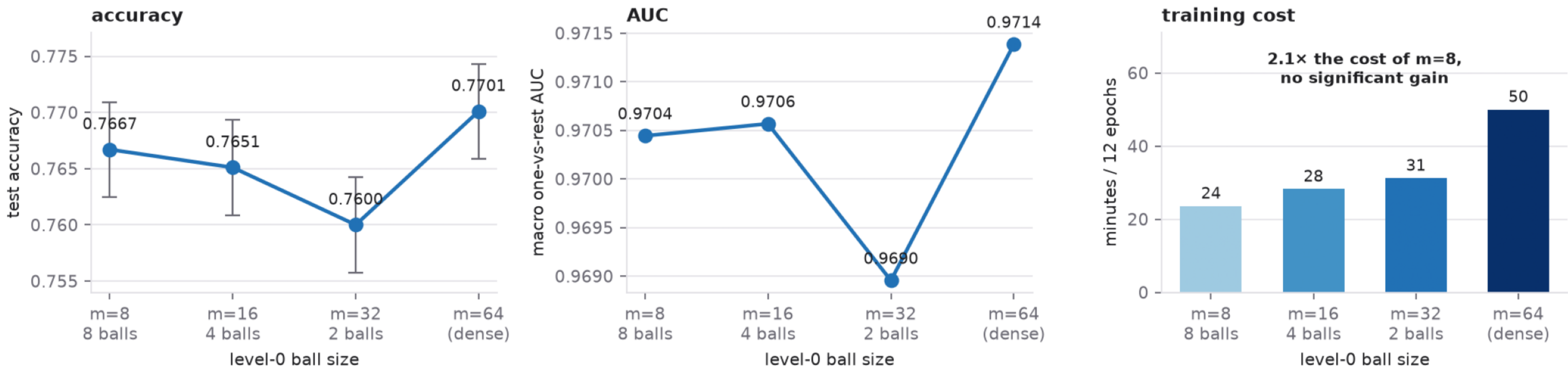
untruncated: identical, p = 0.971

truncated: +3.7 pts, p = 0.013

overall p = 0.449

Is ball locality costing us accuracy?

S11 · level-0 ball size · 80k train / 10k test · 12 epochs · 2.21M params at every m



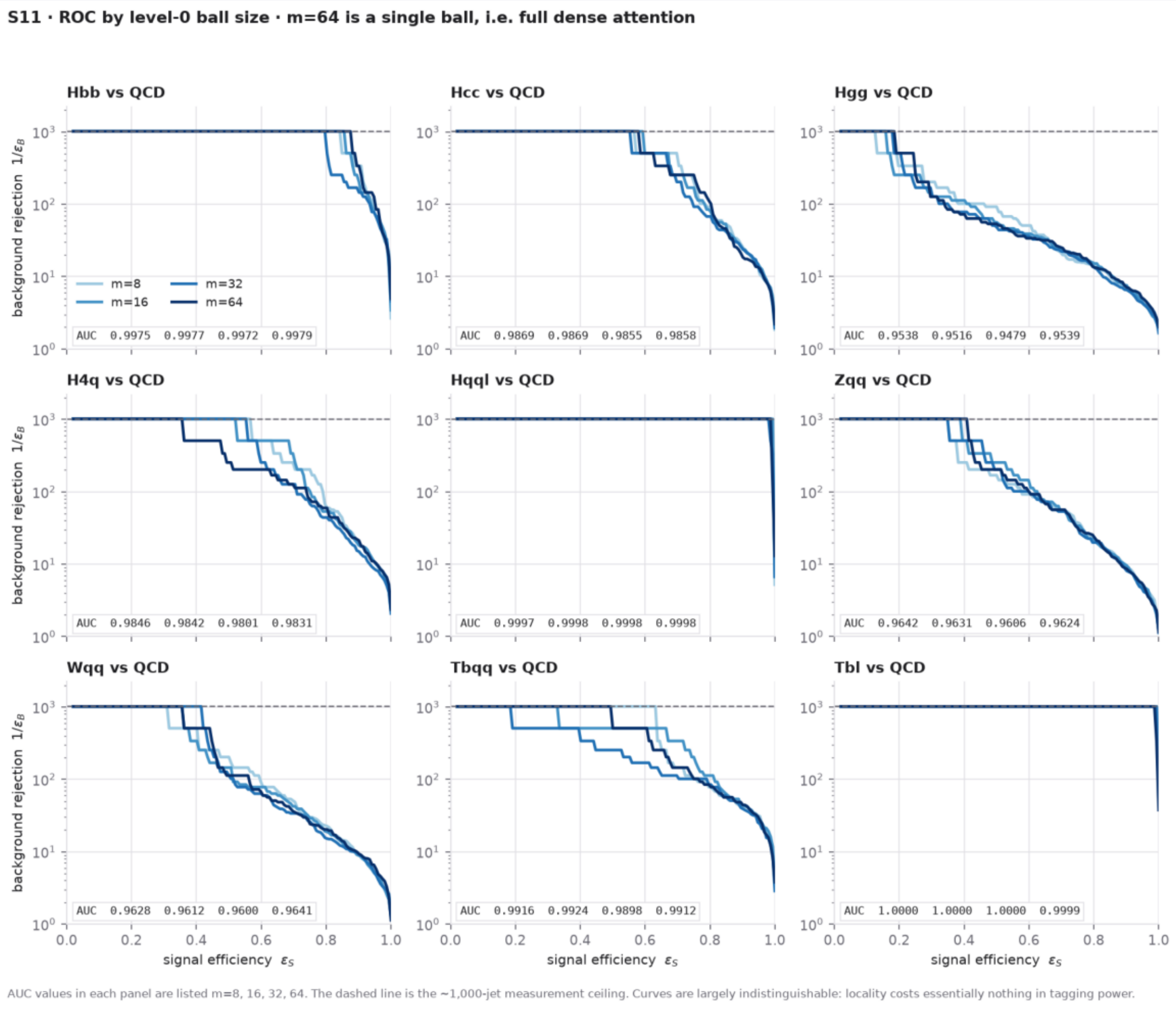
Error bars are binomial on 10,000 test jets. Paired McNemar: m=64 vs m=8 p=0.295 (not significant); m=32 is the only arm that differs significantly from the rest (p=0.044 vs m=8, p=0.002 vs m=64).

m=64 IS dense attention

vs m=8: p = 0.295, not significant

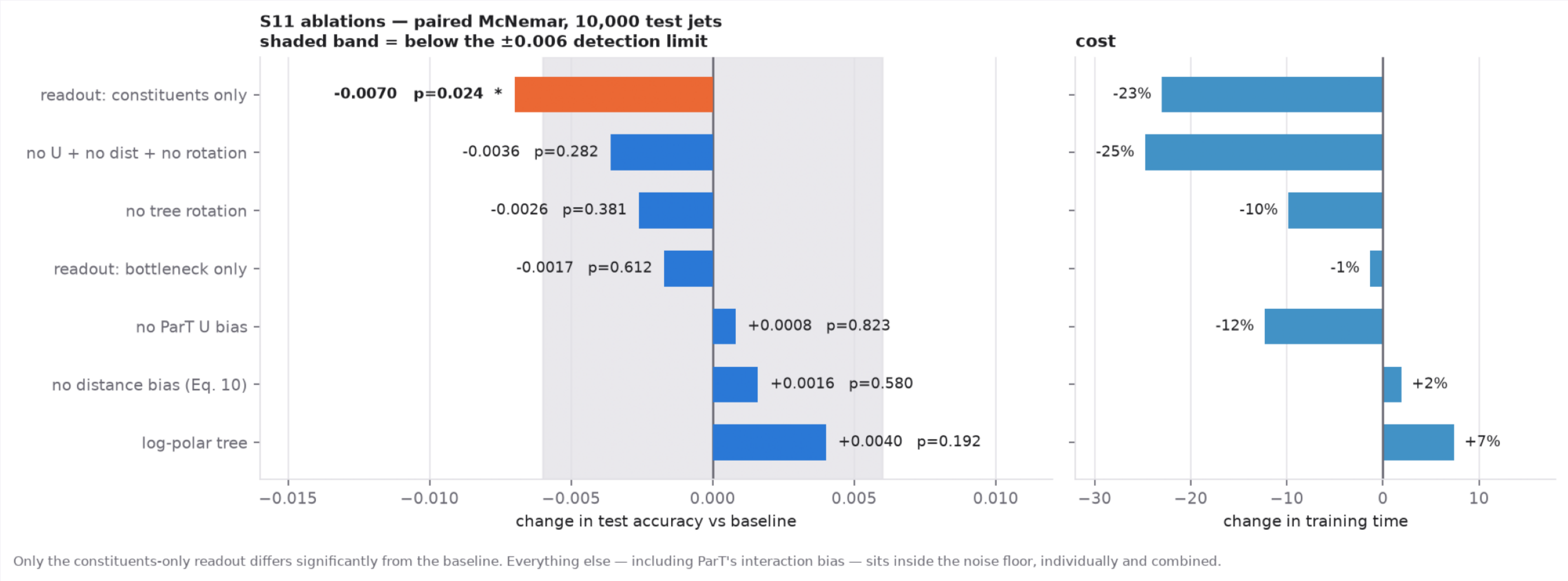
2.1x the training time

Does that hide a class-specific effect?



curves overlap in all nine classes

Which parts of the architecture actually matter?

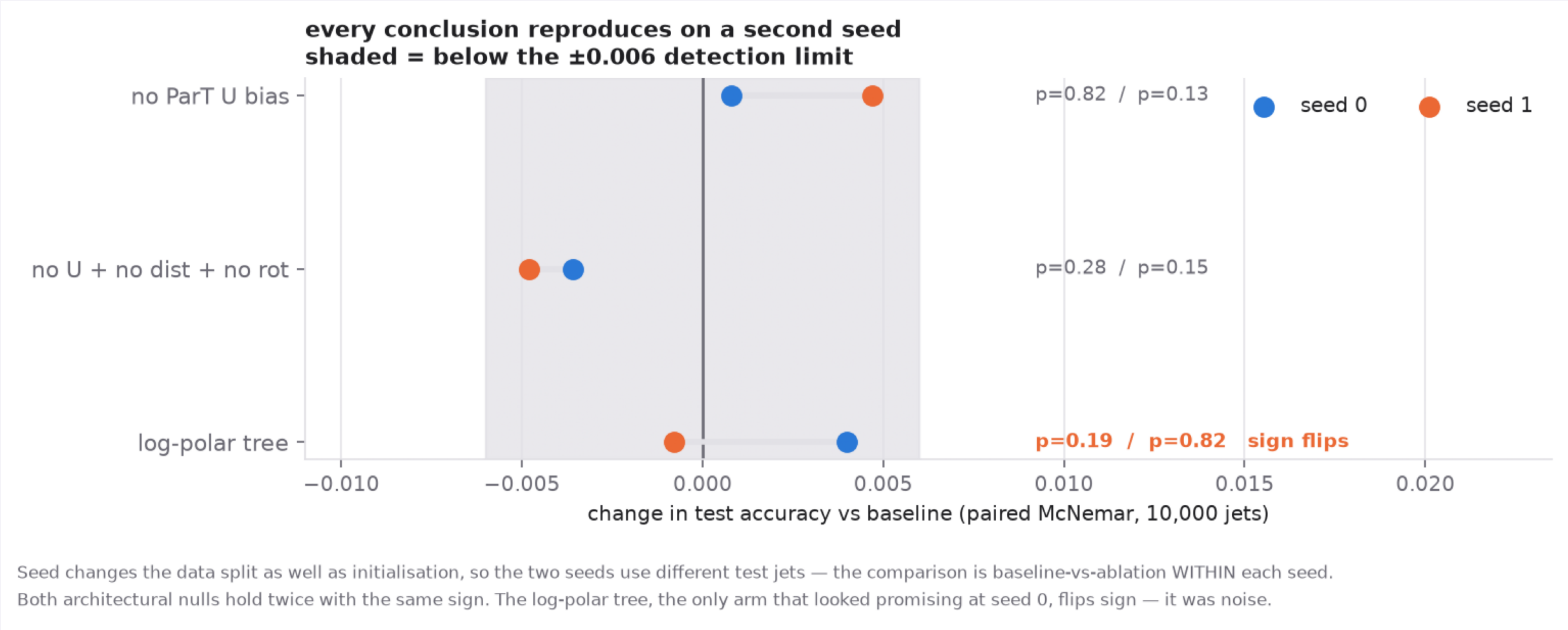


ParT's U bias: $p = 0.823$

only the readout is significant

detection limit ± 0.006

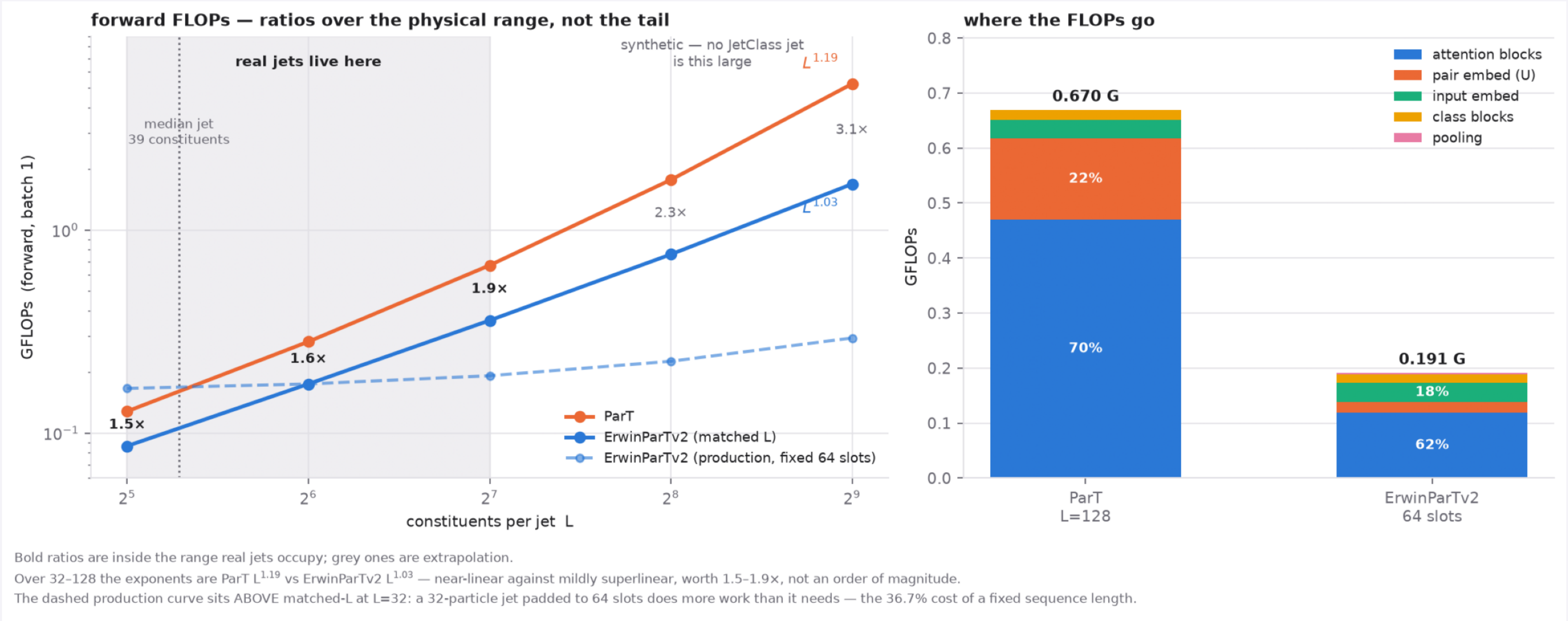
The same conclusions on a second seed



both nulls hold twice

log-polar flips sign - it was noise

Efficiency, scoped to jets that actually exist



1.5-1.9x over the physical range

$L^{1.03}$ vs $L^{1.19}$

$L > 128$ is synthetic

What the data do and do not establish

SUPPORTED

- Dense attention is not significantly better than 8-particle balls (p = 0.295)
- The hierarchy is load-bearing; the readout must see coarse levels
- U bias, distance bias and rotation are all removable — confirmed on two seeds
- 1.5–1.9× fewer FLOPs; trains where ParT cannot on 16 GB
- L=64 costs 0.115% of jet pT; effect confined to 6.3% of jets

NOT SUPPORTED

- That it tags better than ParT — no baseline exists locally
- That coarse-level pair features help — predicted, measured, did not
- Effects smaller than ±0.006 accuracy — below our detection limit
- Anything at 100M-jet scale; 80k jets is 0.08% of JetClass
- That the stripped model should ship — the nulls are bounded, not zero

Where this leaves us

1

Locality is not costing us accuracy.

Dense attention inside our own hierarchy is no better than 8-particle balls, $p = 0.295$.

2

The hierarchy is what matters.

The readout must see coarse levels; everything else we inherited is removable, on two seeds.

3

The efficiency is real but modest at jet scale.

1.5–1.9× FLOPs — and the difference between training and not training on 16 GB.

4

We cannot yet say it tags better.

No local ParT baseline exists. That is the open question, not a footnote.

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Next: the cluster run.

kube/erwinpartv2-job.yml, full JetClass, cheap configuration first.